

Session 2 · Claude Code as Your Coding Copilot

One question, answered with code: what is Apple worth? You will fetch the peers' filings live from the U.S. Securities and Exchange Commission (SEC), compute what their prices imply, compare that with what the market charges, and publish the tool that does it.

Plan: concepts 15' · live demo 22' · your lab 30' · debrief 8'

By the end of Session 2 you can

1. **Drive** Claude Code professionally: one function at a time, reviewed
2. **Fetch** company fundamentals live from EDGAR, the SEC's public filing database
3. **Compute** what the market charges per unit of earnings, across a peer group
4. **Value** Apple at its peers' ratings — as a point, then as a range
5. **Publish** the finished tool to your own GitHub

What we are doing, and why

- **The market gives every company a price**, but a price is not a judgment: it tells you what buyers pay, not whether they should.
This is **the market price** — the latest close.
- **We compute an independent estimate of worth** from public SEC filings.
This is **the implied price**: the peers' ratings — price per unit of earnings — applied to the company's own earnings.
- **The difference between the two is the product**. It does not say "buy" or "sell"; it identifies the question to investigate next.
This is **the premium** (or discount) — measured, not judged.
- **We build the analysis as a tool, with Claude Code, and publish it**.
A reproducible analysis can be rerun, audited and extended; and directing and reviewing AI-generated code with discipline is a core professional

You are the analyst in charge: three habits

- **Ask small:** request one function at a time, so every change is reviewable
- **Read before you run:** accept only changes you can explain
- **Commit at every green moment:** small commits make each step reversible

Pandas in one schema

`DataFrame` = your spreadsheet, driven by commands

`df["ev_m"] / df["ebitda_m"]` → the multiple for all companies at once

- **Operations apply to entire columns at once**, which is why the lab needs no loops
- **No one is expected to memorize pandas**: you specify, the panel writes, you verify

The storyline of this session

the peers' SEC filings



growth and margins



the multiples



Apple valued at the median



the valuation range



your GitHub

fetches live from EDGAR, the SEC's database

Exercise 1 · know the peer group

Exercise 2 · what the market charges

Part B · the implied price vs the market's

Exercise 4 · every peer rating applied

Part C · the tool, published

The vocabulary of comparable-company analysis

Comparable-company analysis values a company *relative to its peers*:
what do their prices imply this one is worth?

Term	Formula	What it measures
EBITDA — earnings before interest, taxes, depreciation & amortization	operating income + D&A	operating earnings, before financing and accounting choices
Market capitalization	shares × share price	the price of the equity
Enterprise value (EV)	market cap + debt – cash	the price of the whole business

The two multiples we use

A **multiple** is a **rating**, not a **price**: it becomes a price estimate only when applied to a company's own earnings. It divides what you pay by what you get:

Term	Formula	What it measures
EV/EBITDA	enterprise value ÷ EBITDA	the whole business over what it earns
Price-to-earnings (P/E)	market cap ÷ net income	the equity per unit of net profit

- You compute both; the valuation uses **EV/EBITDA**, because it compares whole businesses before financing choices distort the picture
- **P/E is reported alongside** — it is the multiple the press quotes, and

Pricing a company with peer multiples

The chain from a peer multiple to a share price:

1. peer EV/EBITDA multiple $\times$ company's EBITDA = implied enterprise value
2. implied EV $-$ debt + cash = implied equity value
3. implied equity value $\div$ shares = implied share price

- **Peers are a choice:** the dataset holds Apple and seven peers whose current earnings are normal — a rating is only fair when its earnings are
- **The deterministic version** applies one rating, the peer *median*, and produces one implied price to set against the market's
- **The range version (Exercise 4)** applies *every* peer rating — where the market price sits among the seven implied prices is the finding

The libraries in this session

Library	Role here	Standing
<code>pandas</code>	tables, cleaning, ratios	the industry standard for data work
<code>plotly</code>	the interactive charts	best in class for interactive visuals
<code>numpy</code>	vectorized arithmetic	the numerical foundation of Python
<code>matplotlib</code> / <code>seaborn</code>	static charts	the long-standing defaults
<code>git</code> + GitHub	version and publish	universal in professional work

How the labs work

Every exercise sits between two markers. **You fill the gaps. Nothing else changes.**

```
### START CODE HERE ###  
df["revenue_growth_1y"] = df[None] / df[None] - 1  
### END CODE HERE ###
```

- **None** → replace with the correct column, value or variable
- **[QUESTION IN CAPITALS]** → replace with the text the bracket asks for
- **Everything else is given.** Do not rewrite it.
- Then run the  **check cell** directly below. Green means correct: continue.

Stuck for two minutes? Select the lines, press **Option+K** (**Alt+K** on Windows), and ask the ***** panel.

Your lab · notebook 02 · 30 minutes

- **The opening cell** fetches the eight companies live from SEC EDGAR
- **Exercise 1:** know the peer group — growth and margins
- **Exercise 2:** the EV bridge and the multiples
- **Exercise 3:** the summary table whose **median** anchors the valuation
- **Part B:** read Apple valued end to end, then **Exercise 4:** the same chain at every peer rating — the valuation range
- **To finish:** publish the tool to your own GitHub repository

Key takeaway

Comparable-company analysis gives every company a peer-implied value range.

Analysis

begins where the market price and that range disagree.

Next session: debugging a valuation model with deliberate errors.